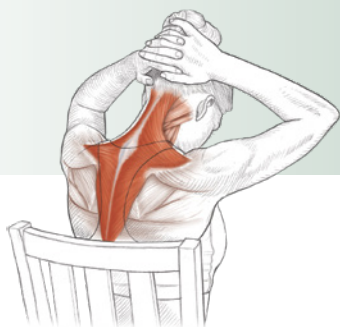




NECK

The seven cervical vertebrae along with associated muscles and ligaments make up the flexible framework of the neck. The vertebrae, muscles, and ligaments work together to support and move the head. The first and second cervical vertebrae have unique shapes and are called the atlas and axis. The atlas is a bony ring that supports the skull. The axis has an upward peg-like projection, the dens, that gives the atlas a point to pivot around. The axis and the other five cervical vertebrae have a posterior bony protuberance, or spinous process, that attaches to the large, thick nuchal ligament. As in the trunk, the vertebral bodies (the oval-shaped bone mass) are connected by posterior and anterior ligaments, along with other ligaments that connect each spinous and transverse (lateral bony protuberance) process to their corresponding parts on the adjacent vertebrae. In addition, each vertebra is separated by an intervertebral disc. Through compression of the vertebrae on the discs, the neck can move forward, backward, and sideways.

The neck muscles are located in two triangular regions called the anterior (front) and posterior (back) triangles. The borders of the anterior triangle are the mandible (jawbone), the sternum (breastbone), and the sternocleidomastoid muscle. The major anterior muscles are the sternocleidomastoid and scalene (figure 8.1*a*). The borders of the posterior triangle are the clavicle (collarbone), sternocleidomastoid muscle, and trapezius muscle. The major posterior muscles (figure 8.1*b*) are the trapezius, longissimus capitis, semispinalis capitis, and splenius capitis.

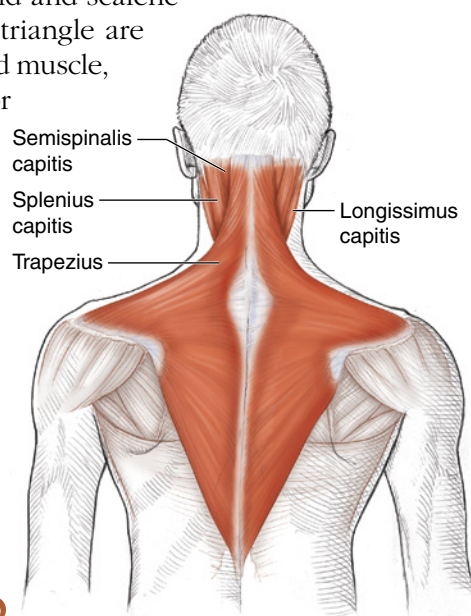
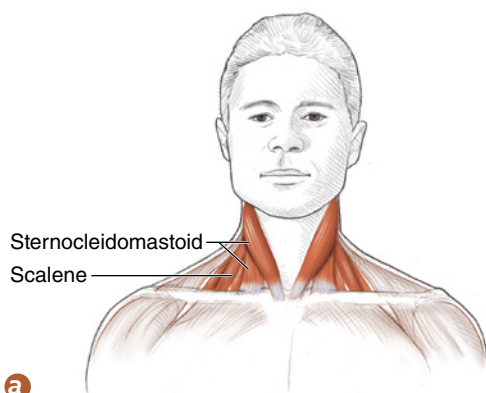


FIGURE 8.1 Neck muscles: (a) anterior; (b) posterior.

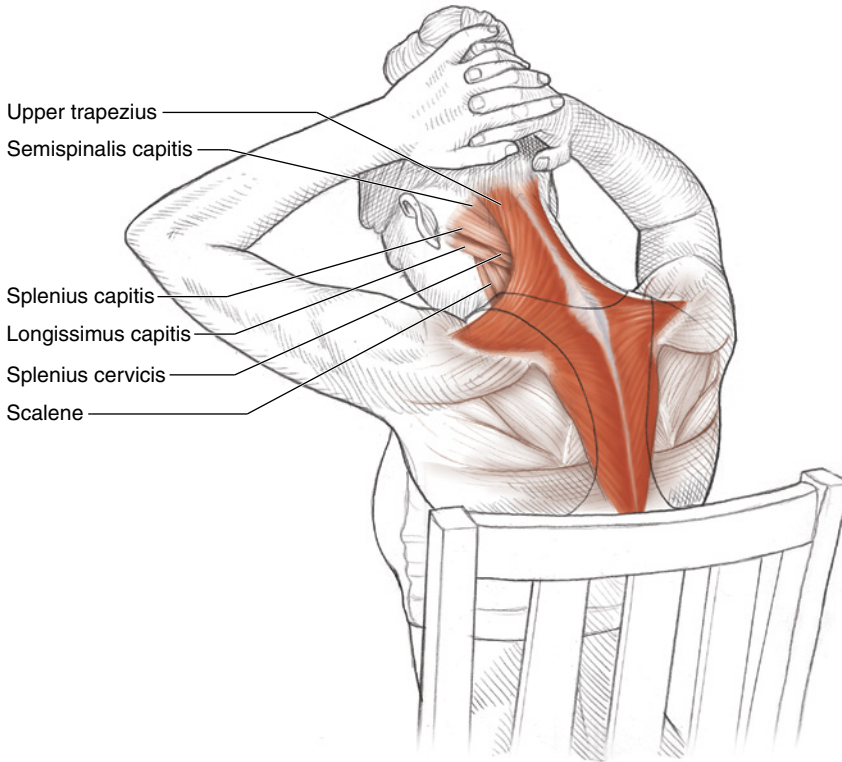
The head movements are flexion (head tilted forward), extension (head tilted backward), lateral flexion and extension (head tipped from side to side), and rotation. Because the muscles in the neck occur in right and left pairings, all the neck muscles are involved in lateral flexion and extension. For example, the right sternocleidomastoid helps perform right lateral flexion, and the left sternocleidomastoid helps perform right lateral extension. Neck flexion is limited not only by the stiffness of the posterior muscles but also by the stiffness of the posterior ligaments, the strength of the flexor muscles, the alignment of the vertebral bodies with the adjacent vertebrae, the compressibility of the anterior portions of the intervertebral discs, and the contact of the chin with the chest. Similarly, neck extension is controlled by the stiffness of the anterior muscles as well as by the stiffness of the anterior ligaments, the strength of the extensor muscles, the alignment of the vertebral bodies with the adjacent vertebrae, and the compressibility of the posterior portions of the intervertebral discs. Finally, in addition to the stiffness of the contralateral muscles and tendons, neck lateral function is controlled by the impingement of each vertebra's transverse process on the adjacent transverse process.

People seldom consider the neck muscles when stretching. Neck flexibility probably does not cross your mind until you discover that you have a stiff neck. A stiff neck is commonly associated with sleeping in a strange position (such as on a long flight) or sitting at a desk for an extended time, but a stiff neck can result from almost any type of physical activity. This is especially true for activities in which the head must be held in a constantly stable position. A stiff neck can also have a negative effect in sports in which head position is important, such as golf, or when rapid head movements are important for tracking the flight of an object, such as in racket sports. Poor neck flexibility usually results from holding the head in the same position for long periods. In addition, a fatigued neck muscle can stiffen up after exercise. The exercises in this chapter can help keep the neck from stiffening up after exercises, unusual postures, or awkward sleep positions.

Because all the major muscles in the neck are involved in neck rotation, it is fairly easy to stretch the neck muscles. The first consideration when choosing a particular neck stretch should be whether greater stiffness occurs with flexion or extension. Therefore, the first exercise in the two exercise groupings focuses on these specific actions. Once you achieve greater flexibility in either pure flexion or pure extension, then you can add a stretch that includes lateral movement. In other words, to increase the flexibility of the neck extensors, start with the neck extensor stretch and then, as flexibility increases, add the neck extensor and rotation stretch.

Stretching the neck can be dangerous if not done properly. Some stretches of the neck use what is termed a plow position in which the back of the head lies on a surface, with the trunk nearly perpendicular. This position can generate high stress at the bending point, especially in people with low neck flexibility. This high stress can either damage the vertebrae or greatly compress the anterior intervertebral disc. Disc compression can cause protrusion and pressure on the spinal cord, thus damaging it. Additionally, when stretching the neck, be careful to avoid applying sudden or rapid force. Sudden force application can lead to whiplash injuries; in the worst-case scenario, whiplash can sever the vertebral arteries and force the dens into the brain's medulla oblongata, causing death.

NECK EXTENSOR STRETCH



Execution

1. Sit comfortably with the back straight.
2. Interlock the hands on the back of the head near the crown.
3. Lightly pull the head straight down and try to touch the chin to the chest.

Muscles Stretched

Most-stretched muscle: Upper trapezius

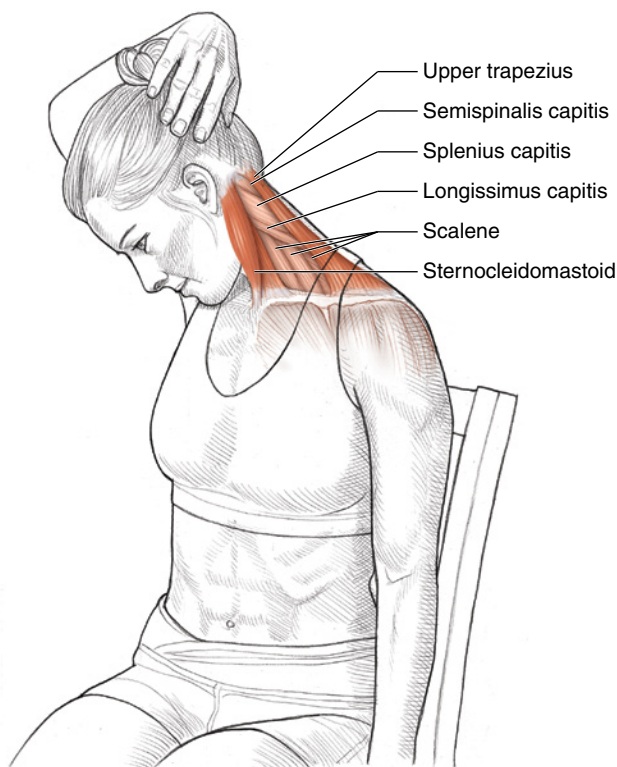
Less-stretched muscles: Longissimus capitis, semispinalis capitis, splenius capitis, splenius cervicis, scalene

Stretch Notes

You can do this stretch while either sitting or standing. A greater stretch is applied when seated. Standing reduces the ability to stretch because reflexes come into play to prevent a loss of balance. Therefore, we recommend doing the stretch while seated. During the stretch, make sure the shoulders stay down. Hunching the shoulders reduces the stretch. Also, keep the neck as straight as possible (no curving). Try to touch the chin to the lowest possible point on the chest.

It is common for people who are stressed to hunch their shoulders. Constantly hunching limits the ability of the posterior neck muscles to relax. This causes these muscles to become tight, adding to the pain and fatigue and causing more hunching. Additionally, these muscles can become tight after neck strain or a whiplash injury. Relief and relaxation can be obtained by doing this stretch, thus greatly decreasing hunching. Also, the neck extensor muscles must remain loose in order to maintain proper posture, and maintaining proper posture can in turn help reduce muscle strain and tightness.

NECK EXTENSOR AND ROTATION STRETCH



Execution

1. Sit comfortably with the back straight.
2. Place the right hand on the back of the head near the crown.
3. Pull the head down and to the right so that it points to the right shoulder. Bring the chin as close to the right shoulder as possible.
4. Repeat the stretch on the other side.

Muscles Stretched

Most-stretched muscles: Left upper trapezius, left sternocleidomastoid

Less-stretched muscles: Left longissimus capitis, left semispinalis capitis, left splenius capitis, left scalene

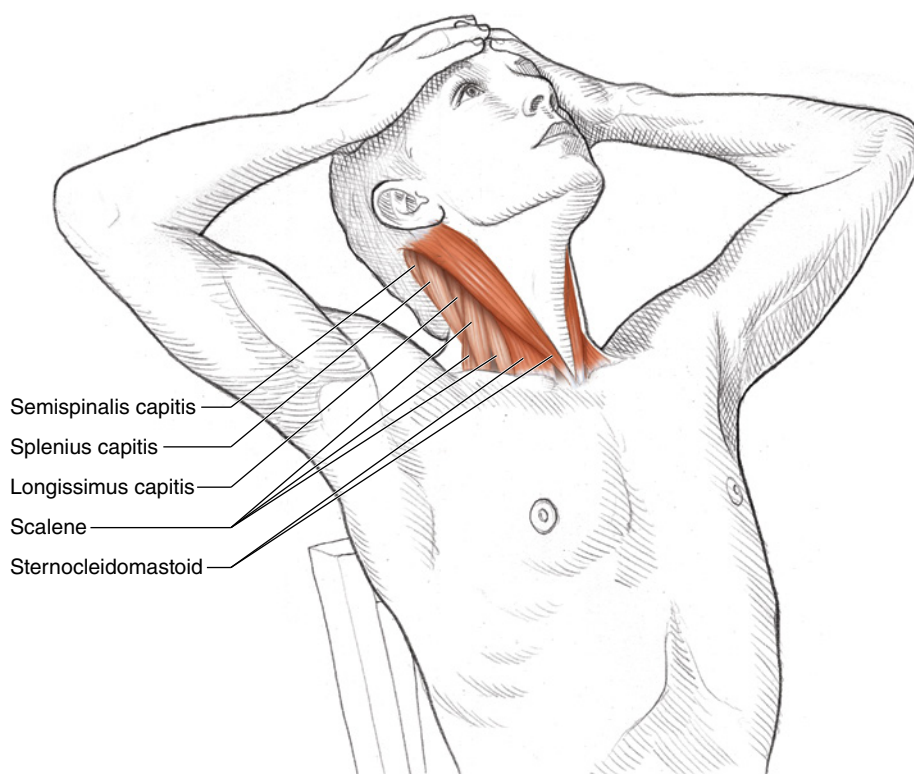
Stretch Notes

After the neck extensors become flexible, you can progress from stretching both sides of the neck simultaneously to stretching the left and right sides individually. Stretching one side at a time allows you to place a greater stretch on the muscles. Often one side of the neck is stiffer than the other side. Frequently this occurs if you sleep strictly on one side or sit at a desk and do not look straight ahead but continually look either to the left or the right.

When you stretch both sides of the neck simultaneously, the amount of stretch applied is limited by the stiffest muscles. Thus, if one side is more flexible, it may not receive a sufficient stretch. By stretching each side individually, you can concentrate more effort on the stiffer side.

You can perform this stretch while either sitting or standing. Although you can achieve a better stretch while sitting, do whichever feels best to you.

NECK FLEXOR STRETCH



Execution

1. Sit comfortably with the back straight.
2. Interlock the hands and place the palms on the forehead.
3. Pull the head back so that the nose points straight up to the ceiling.

Muscles Stretched

Most-stretched muscle: Sternocleidomastoid

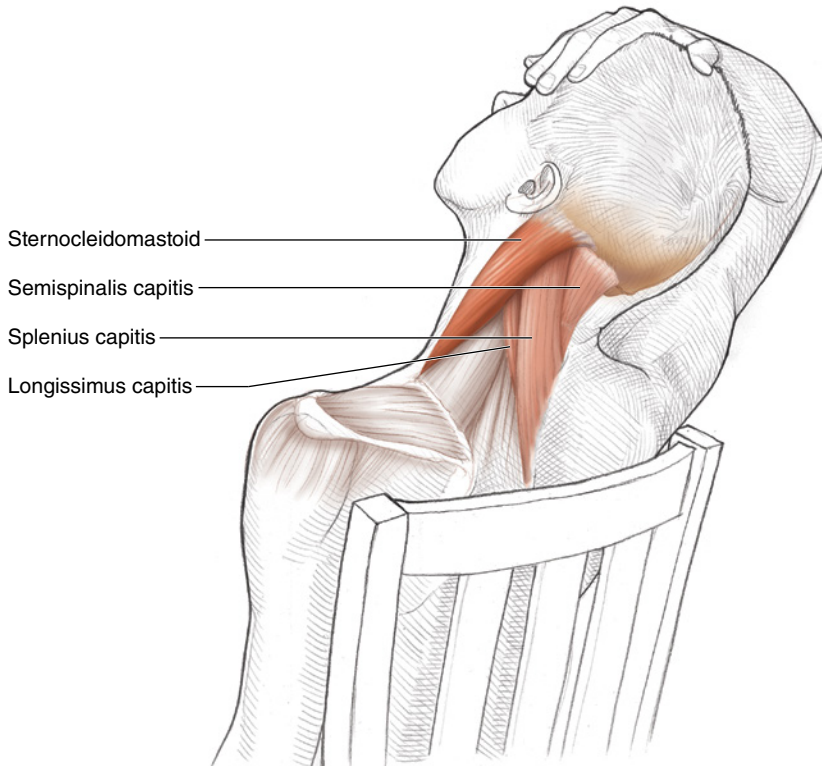
Less-stretched muscles: Longissimus capitis, semispinalis capitis, splenius capitis, scalene

Stretch Notes

You can do this stretch while either sitting or standing. A greater stretch is applied when seated. Standing reduces the ability to stretch because reflexes come into play to prevent a loss of balance. Therefore, we recommend doing the stretch while seated. During the stretch, keep the shoulders down. Hunching the shoulders reduces the stretch. Also try to bring the chin as far back as possible.

When people are under stress, they typically breathe forcefully while keeping their shoulders raised. This can lead to pain and tension in the anterior neck muscles. Short-term relief can be obtained by doing this stretch. Also, the neck flexor muscles must remain loose in order to maintain proper posture. If you let these muscles become tight, you can end up with the deformation commonly called “vulture neck,” in which the head position looks like the protruding head of a vulture. To help maintain correct posture, this stretch should be done several times a week.

NECK FLEXOR AND ROTATION STRETCH



Execution

1. Sit comfortably with the back straight.
2. Place the right hand on the forehead.
3. Pull the head back and toward the right so that the head points toward the right shoulder.
4. Repeat for the left side.

Muscles Stretched

Most-stretched muscle: Left sternocleidomastoid

Less-stretched muscles: Left longissimus capitis, left semispinalis capitis, left splenius capitis

Stretch Notes

After the neck flexors become flexible, progress from stretching both sides of the neck simultaneously to stretching the left and right sides individually. Stretching one side at a time allows you to place a greater stretch on the muscles. This is especially important for people who stand hunched over with the head pointed mainly to one side.

When you stretch both sides of the neck simultaneously, the amount of stretch applied is limited by the stiffest muscles. Thus, the more flexible side may not receive a sufficient stretch. By stretching each side individually, you can concentrate more effort on the stiffer side.

You can perform this stretch while either sitting or standing. Although you can achieve a better stretch while sitting, choose whichever position feels best to you.

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